One engine, two markets: pricing with an AI-assisted workflow

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Rank 3 of 54	0.16458	Rank 2
Week 1 private leaderboard, pooled Tweedie D2 0.13864	Week 2 public score of the final ensemble (first file 0.16077)	Week 2 private leaderboard

Week 1: developed market, 2.4 million policy rows.

Week 2: emerging market, two years of data, 76% new business in the test book.

The emerging data decides how much; the developed data decides who.

What changed in the new market

+34% / -42%

motor claim frequency / mean claim cost

76%

of the 2025 test book is new business (14% in training)

2 years

of emerging history, none before 2023

Peril (frequency ratio emerging / developed)	Ratio
glass-hail	1.5 - 1.8
third-party property, bodily injury	1.4 - 1.5
storm	1.3 - 1.7
escape of water	0.65 - 0.7
severity scale (median ratio)	motor 0.6, buildings 0.7, contents 0.75

Ratios are stable between the two emerging years, except for the rare buildings perils (fire 2.3x then 0.8x).

Method

Developed model

2021-24, 2.4M rows, variables both markets record

+ claim history

last year's claims of the policy, address and customer

Ensemble

5 tree models, 7 nets, 4 Tweedie GBMs, equal weights in groups

Level from emerging book

claim counts x winsorised severity per peril, credibility-weighted

Gender factor

one flat factor per line, credibility 300 claims

No emerging-only model: it scores 0.1110 against 0.1405 for the transfer.

Frequency x severity per peril, trended and completeness-adjusted; the direct Tweedie GBM adds ranking, not level.

Level from claim counts, not from a few large claims: closing to the training total copied large-loss noise.

How we trust it

Step	Forward	Reverse
Emerging-only GBM	0.1110	
Developed model + closure + gender	0.1256	0.1162
+ robust level (no closure)	0.1281	0.1176
+ nets	0.1289	0.1220
+ claim history	0.1313	0.1229
+ Tweedie part	0.1336	0.1237
+ buildings credibility 400	0.1397	0.1271
+ ensemble (final)	0.1405	0.1268

Forward: calibrate 2023, score 2024. Reverse: calibrate 2024, score 2023.

Adopt only if neither direction is worse.

Simulation, known truth: robust level beats the line closure in every scenario.

Ceiling: a perfect level adds at most 0.005 D2.

Residual search: no feature structure left after the gender factor (973 levels).

AI in the workflow

Claude Code: data review, test-first code (125 automated tests), experiments, documents.

Helper sessions in separate containers ran the hours-long refits in parallel.

A blind second agent compared the two markets from the raw tables only and found two items we had missed.

Errors the checks caught: three unsupported statements in the first data review; a simulator defect that looked like an estimator bias; a net that Part 1 had already rejected.

Every decision has a numbered row with its evidence (83 so far); every result has a command.

Guardrails and uncertainty

Automated checks on every push; leak alarm (pooled D2 above 0.35 stops a run); reproducibility assertion.

Model card with monitoring triggers (actual / expected by line and peril).

Main stated risk: the buildings level; two years, one large-loss peril swinging 2.3x then 0.8x (+-20% is about +-0.01 D2).

Gender is used as a rating factor and flagged for regulatory review; a gender-free file is the stated alternative.

Office premium: expected claims plus loadings; written premiums roughly equal expected claims, so loadings need about 2.5x the premium.